

Arduino And Javascript

[Click to ★ the Project](#)

WhoAmI

github.com/voidash

Arduino

- A company in Italy
- Arduino is referred to physical programmable circuit board (microcontroller)
- ArduinoIDE

Agenda

Without Arduino Ecosystem

- Tooling is hard as it requires special programming skills and deep knowledge on how chip works and how to communicate with it.
- How to do without Arduino IDE
 - Read the manual
 - While programming disable std features and define the target architecture or download one
 - or program in assembly and use assembler
 - Then use serial interface such as RS-232 to flash the chip.

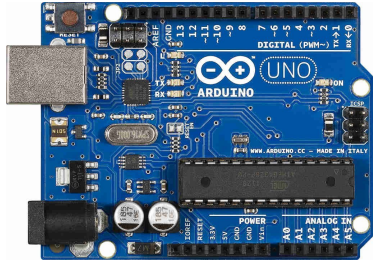
Why Arduino

- Great for prototyping
- cross platform
- Has a very low power requirements
- Is open source with large community
- Doesn't require a degree worth of studying to program it.
- Very Cheap

What is possible with Arduino

- Insert Some projects here

Arduino Flavors

				
Arduino Uno	Arduino Mega	Arduino Nano	Arduino Leonardo	Arduino Lilypad

Getting the Hardware

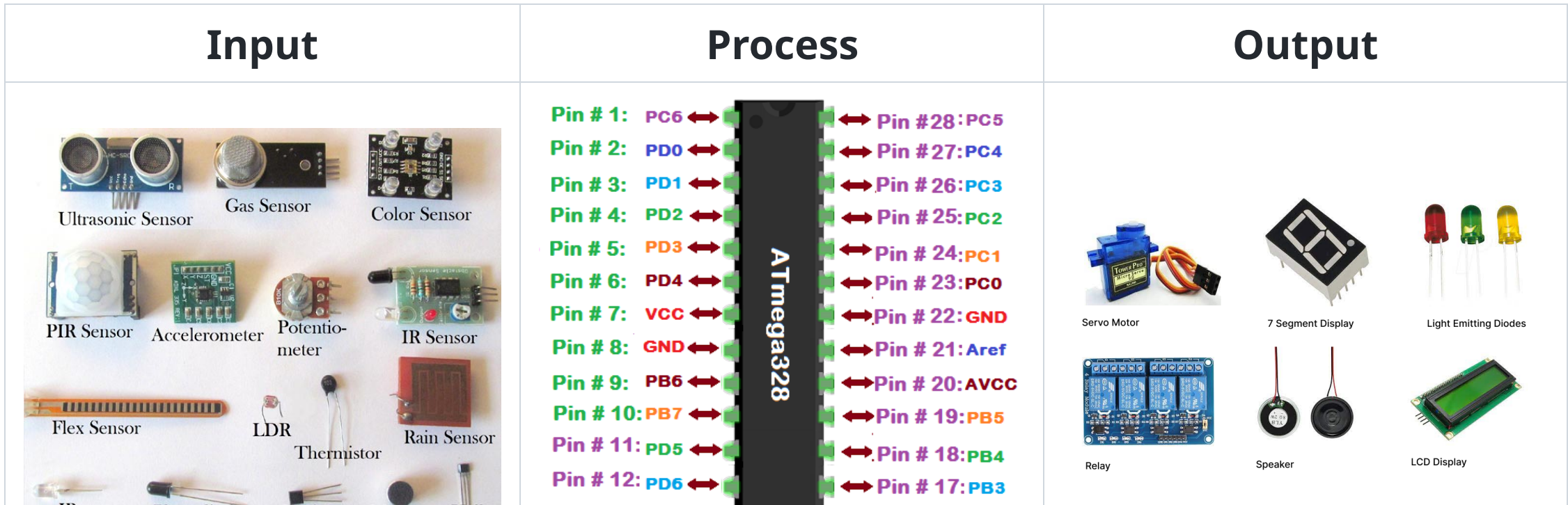
- If you are in Nepal
<https://himalayansolution.com/product/arduino-starter-kit-1>
- Access to International Payment then
<https://www.aliexpress.com/item/1005001653349193.html>

Stores in Nepal

- Supreme Light Technology: <https://www.sltech.com.np/>
- Himalayan Solution <https://himalayansolution.com/>

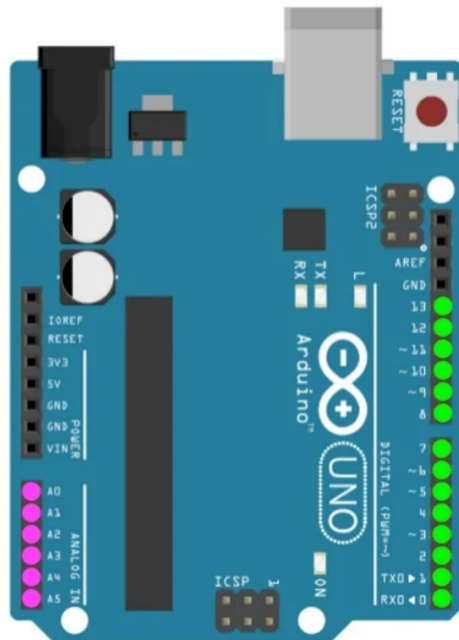
Microcontroller and Microprocessor

- Microcontroller is a chip that houses CPU, Memory(RAM ROM), I/O controllers all in a single package however Microprocessor however only has CPU
- A computer basic operation is `input -> process -> output`
- Atmega328P has 32kb Of program Memory, 1kb of ROM and 2Kb of RAM



Arduino Board

Input

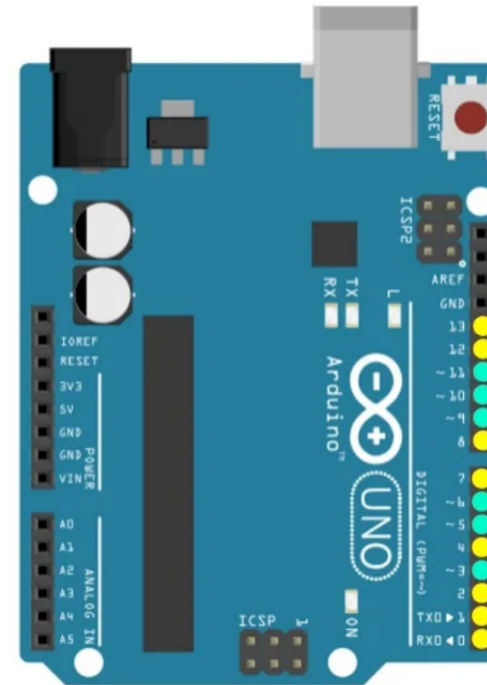


20 Input Pins

14 Digital Input Pins
0-13

6 Analog Input Pins
A0-A5

Output



14 Output Pins

8 0/1 Pins

0,1,2,4,7,8,12,13

6 PWM Pins

3,5,6,9,10,11

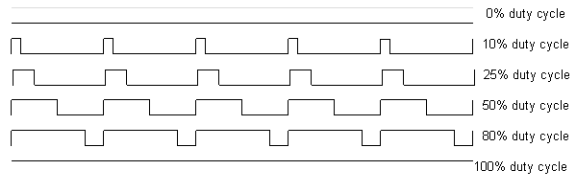
Arduino Board Continued

For Input

- Analog Pins : Used to read analog sensor data. i.e Potentiometer, Pressure Sensor, Accelerometer, LDR sensor
- Digital Pins : If sensor already has A/D converter or the output of the sensor is discrete then Digital pins are used. i.e Ultrasonic Sensor, Ball Switch Sensor. It can only detect 0V or (3.3V/5V).

For Output

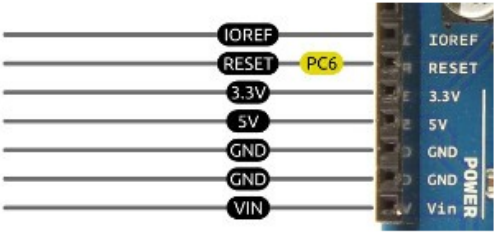
- PWM Pins : Pulse Width Modulation can be thought as a square wave continuously being on and off, where on time is the pulse width and it defines duty cycle.



- Digital Pins: They either provide 0V as off or 5V as On.

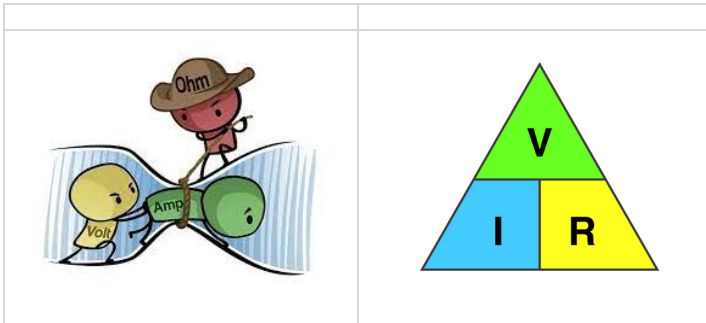
Arduino Board continued

Power Pins

test	ok
 The diagram shows a vertical column of seven pins on the left, each with a label in a black box: IOREF, RESET, 3.3V, 5V, GND, GND, and VIN. To the right of these labels is a small image of the Arduino board's power header, showing the corresponding pins and their labels: IOREF, RESET, 3.3V, 5V, GND, GND, and Vin. A yellow label 'PC6' is placed next to the RESET pin on the left.	• IOREF: Ioreference • RESET: Provide 5V to reset the board • 3.3V: 3.3V output for sensors • 5V: 5V output for sensors • GND: Ground Pin • VIN: Provide 5V to power the Arduino Uno Board

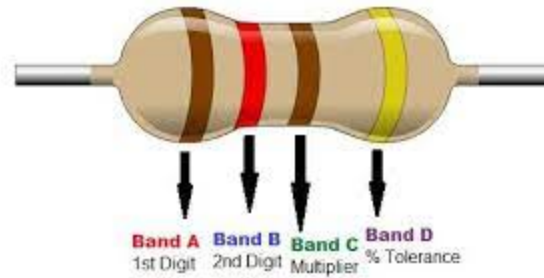
Basics of Electronics

- Ohm's law $V = IR$



- Voltage : Electric Pressure
- Current : Flow of electrons
- Resistance : Measure of opposition of current flow

Resistor

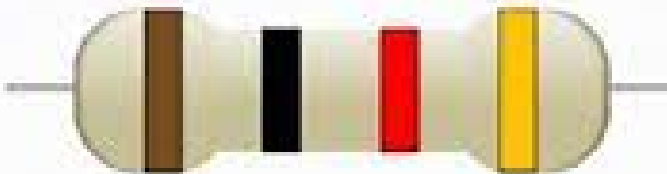


Color	Numeric Value	Multiplier	Tolerance	Temperature coefficient
BLACK	0	1 Ω		250
BROWN	1	10 Ω	$\pm 1\%$	100
RED	2	100 Ω	$\pm 2\%$	50
ORANGE	3	1K Ω		15
YELLOW	4	10 Ω		25
GREEN	5	100 Ω	$\pm 0.5\%$	20
BLUE	6	1M Ω	$\pm 0.25\%$	10
VIOLET	7		$\pm 0.1\%$	5
GREY	8			1
WHITE	9			
GOLD			$\pm 5\%$	
SILVER			$\pm 10\%$	

Exercise

Welcome

Resistor Color Code Practice



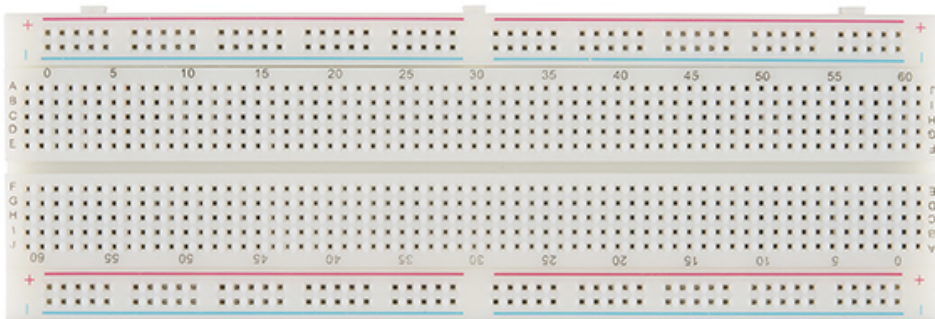


	1 st Digit	2 nd Digit	Multiplier	Tolerance
Black	0	0	x 1	
Brown	1	1	x10	±1%
Red	2	2	x10 ²	±2%
Orange	3	3	x10 ³	±3%
Yellow	4	4	x10 ⁴	±4%
Green	5	5	x10 ⁵	±0.5%
Blue	6	6	x10 ⁶	±0.25%
Violet	7	7	x10 ⁷	±0.1%
Grey	8	8	x10 ⁸	±0.05%
White	9	9	x10 ⁹	
Gold			x10 ⁻¹	±5%
Silver			x10 ⁻²	±10%

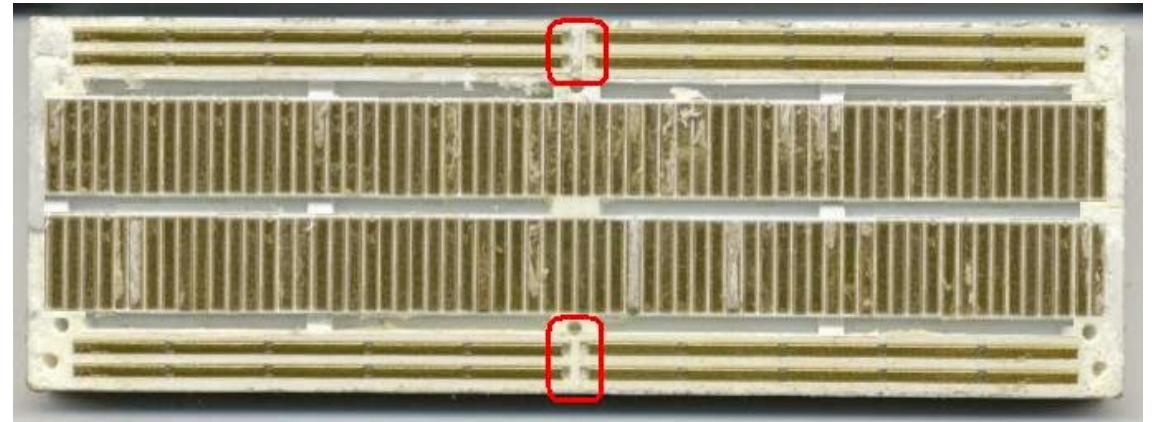
Breadboard

- For Electronics Prototyping

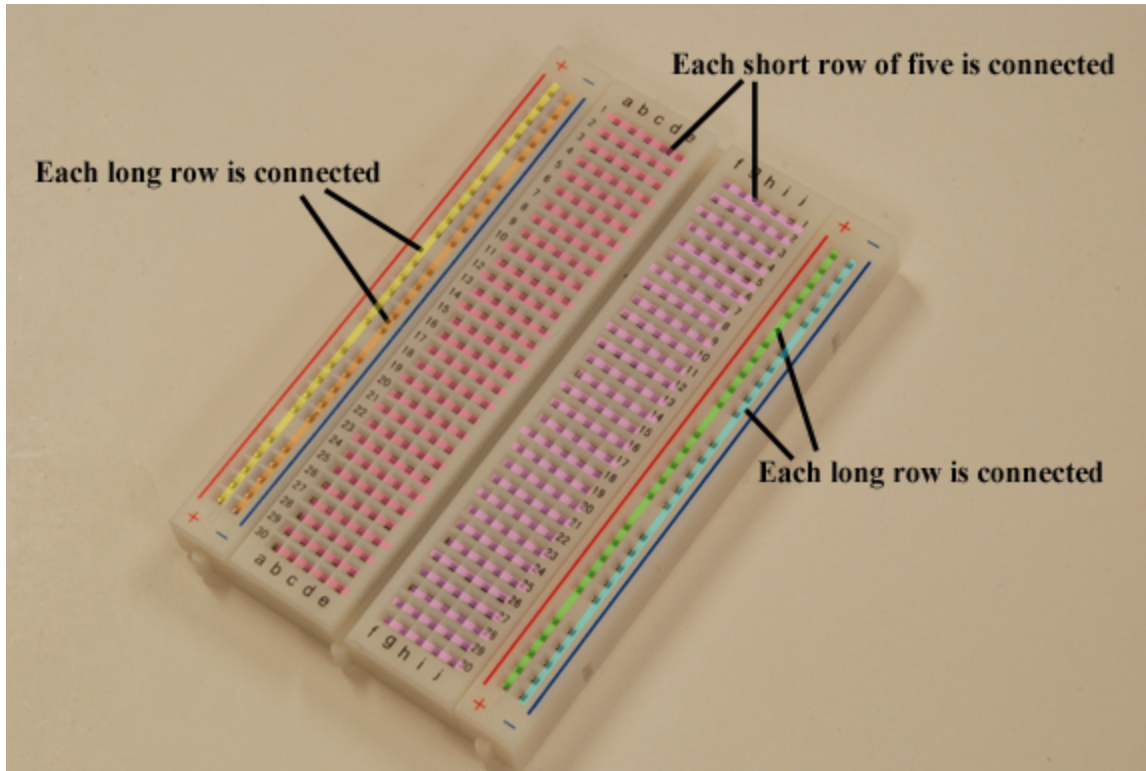
Outside



Inside

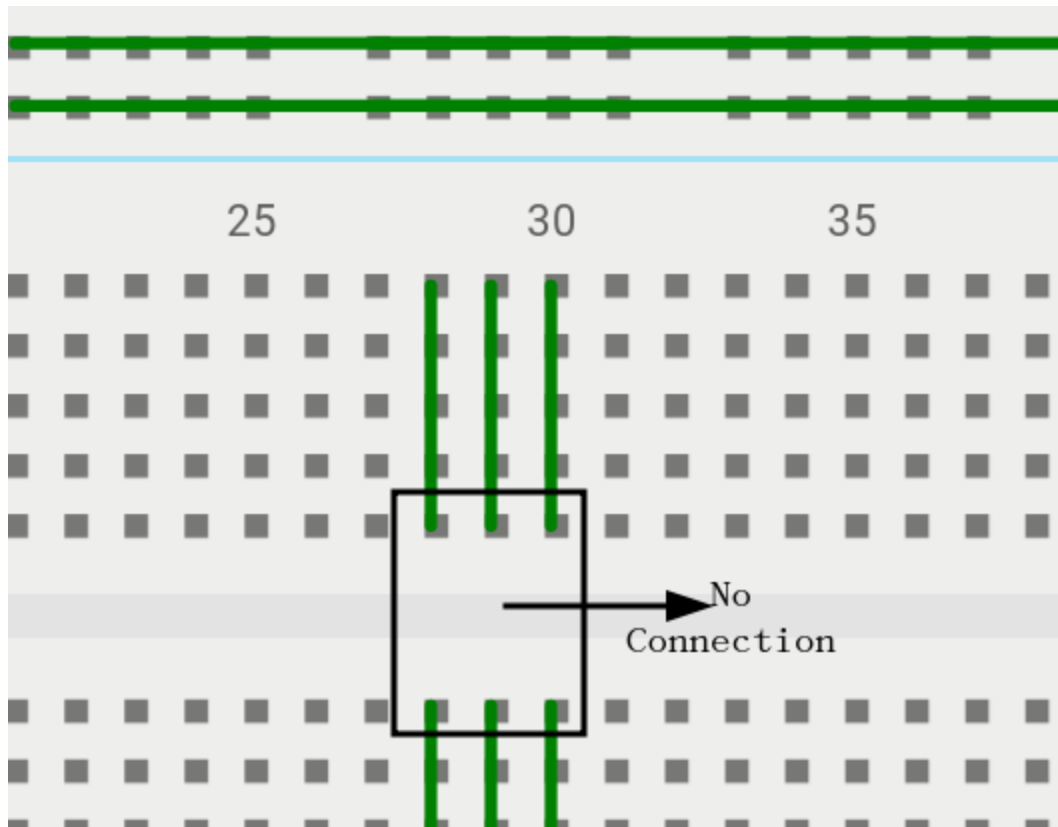


Breadboard Continued

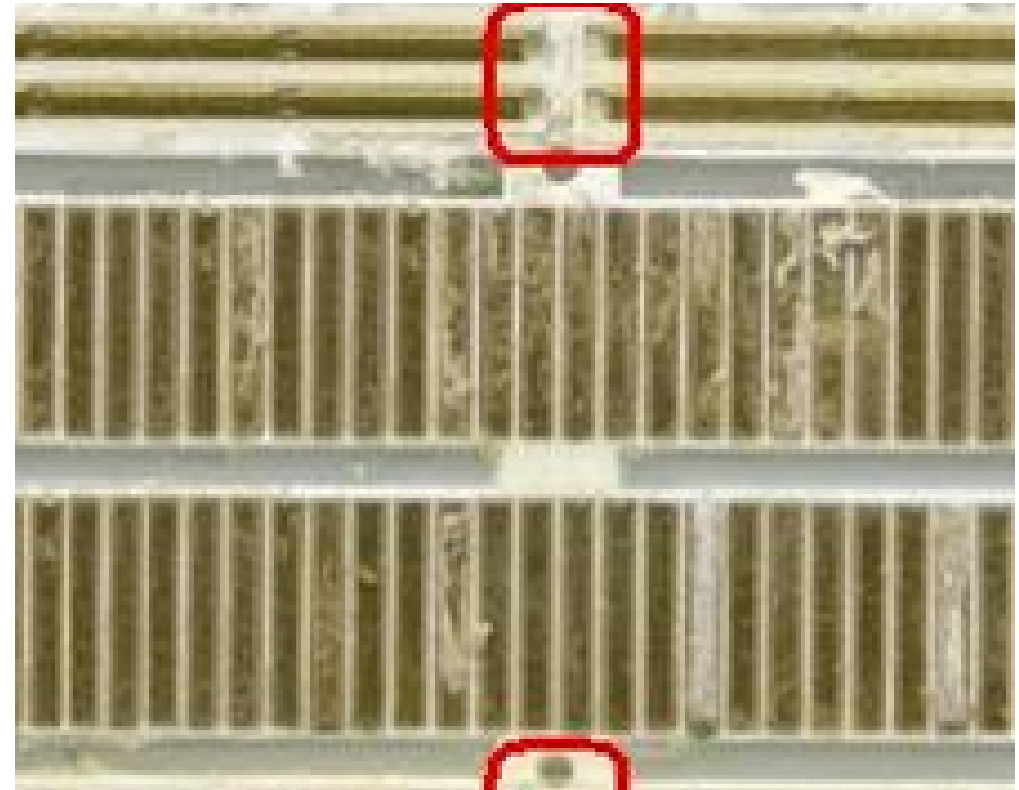


Breadboard Continued

Column1



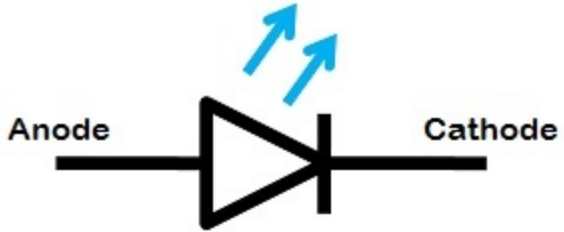
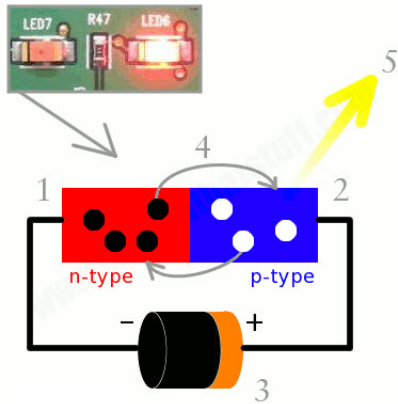
Column2



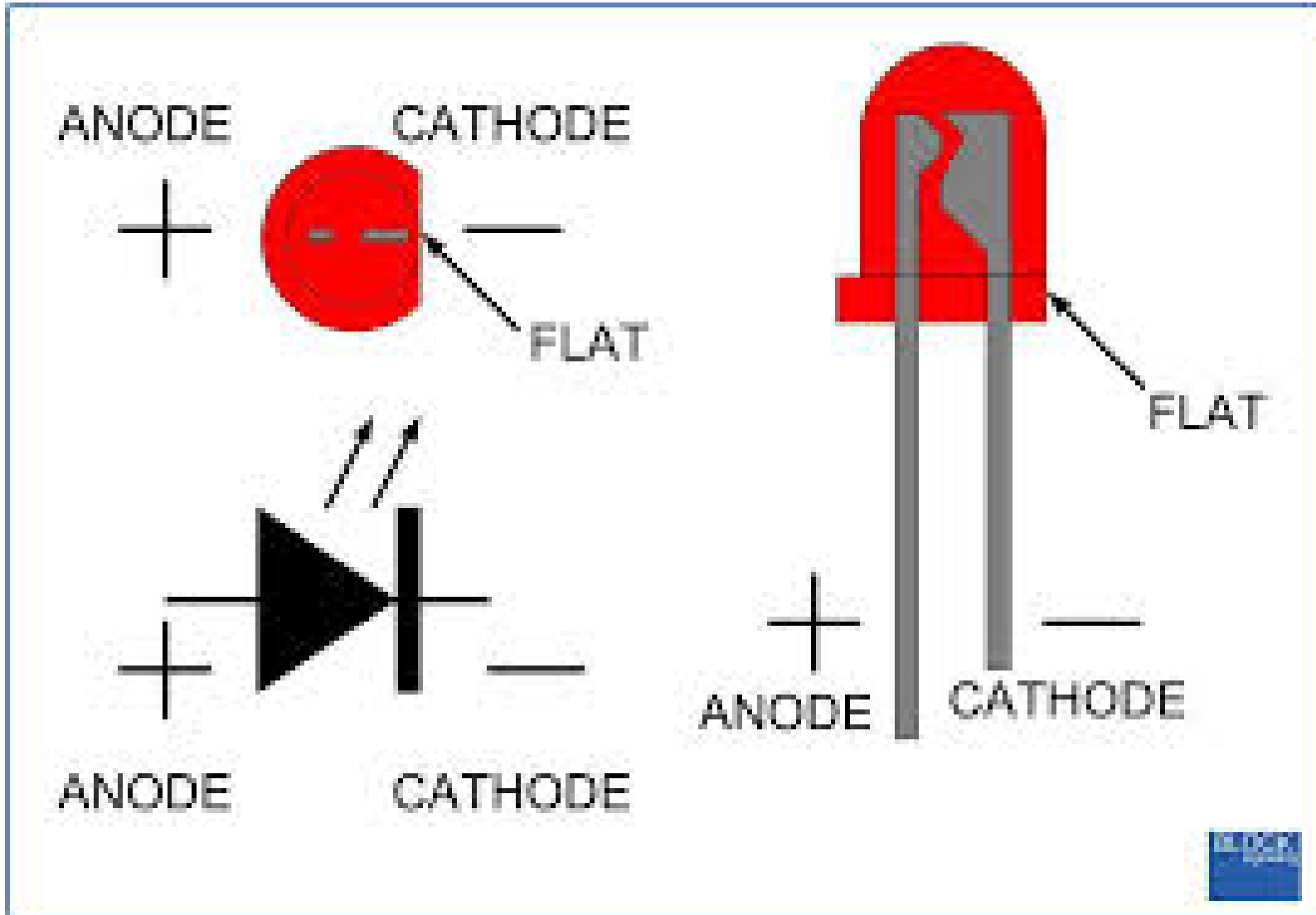
LED (Light Emitting Diode)

<https://www.toppr.com/bytes/principles-of-led/>

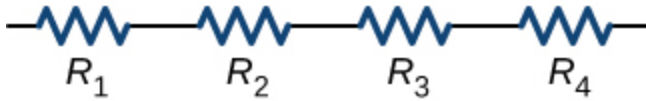
- Active Semiconductor Device

Column1	Column2	Column3
		

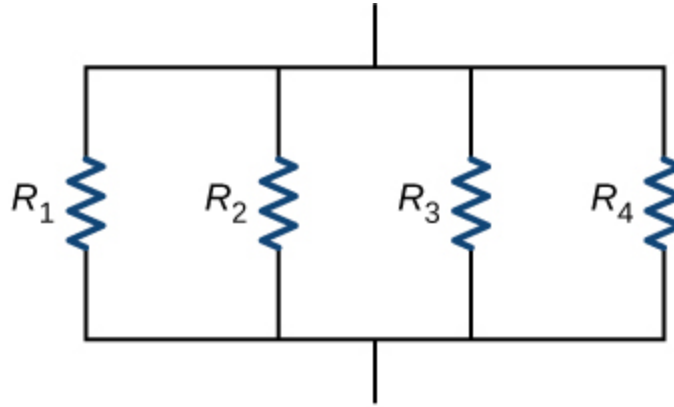
LED



Series and Parallel Resistance



(a) Resistors connected in series



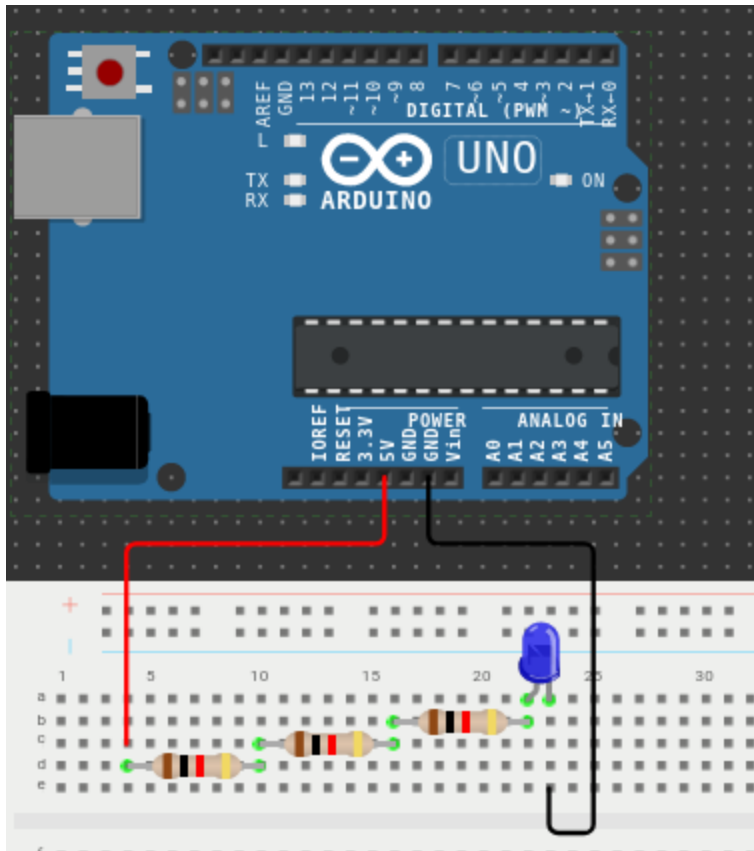
(b) Resistors connected in parallel

- Series Resistance $R_{eq} = R_1 + R_2 + R_3 + R_4$
- Parallel Resistance $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$

Lets Build A Circuit

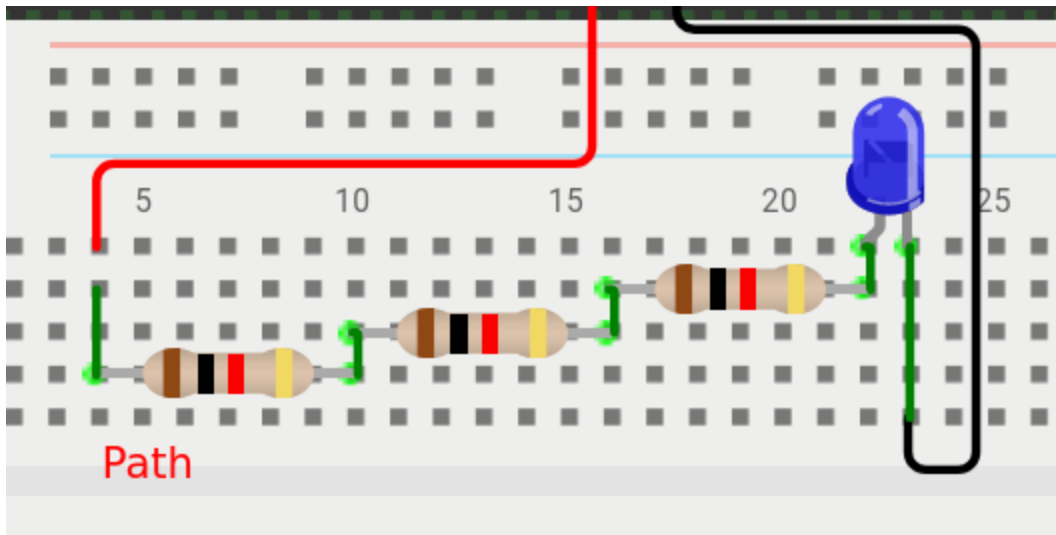
Series Resistance

<https://wokwi.com/projects/376191541801099265>

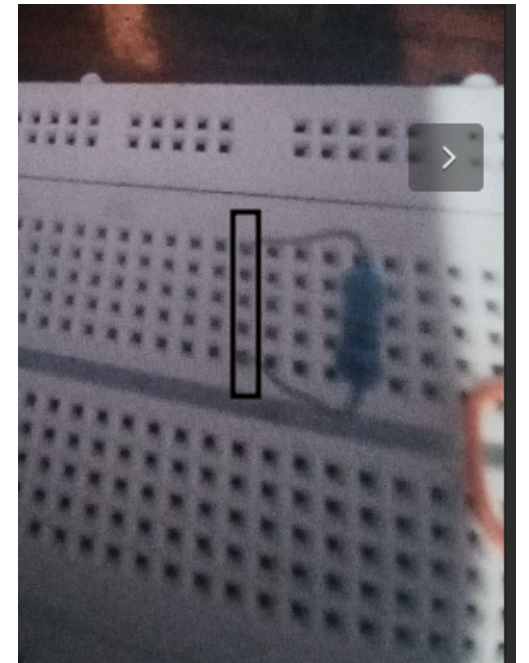
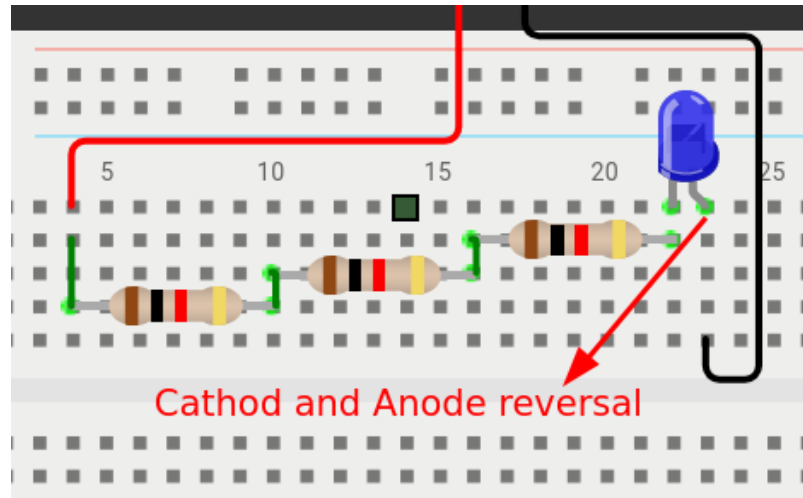
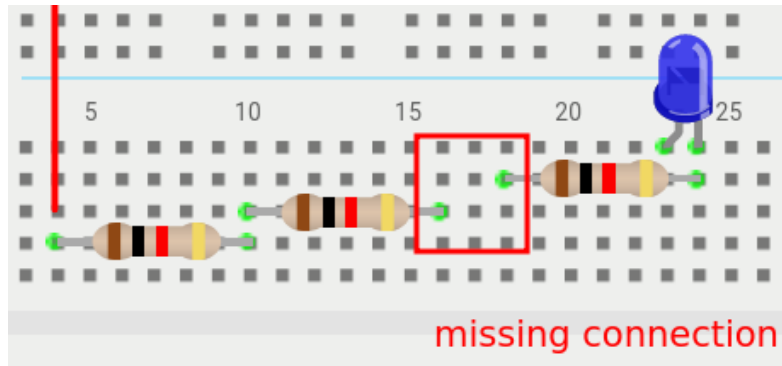


- Was there decrement in LED Glow?

Breadboard Path



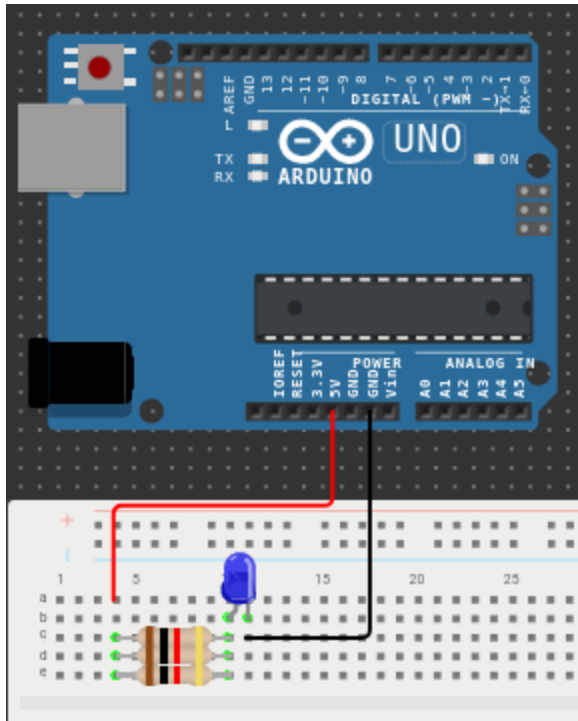
Common Mistakes



Lets Build A Circuit

Parallel Resitance

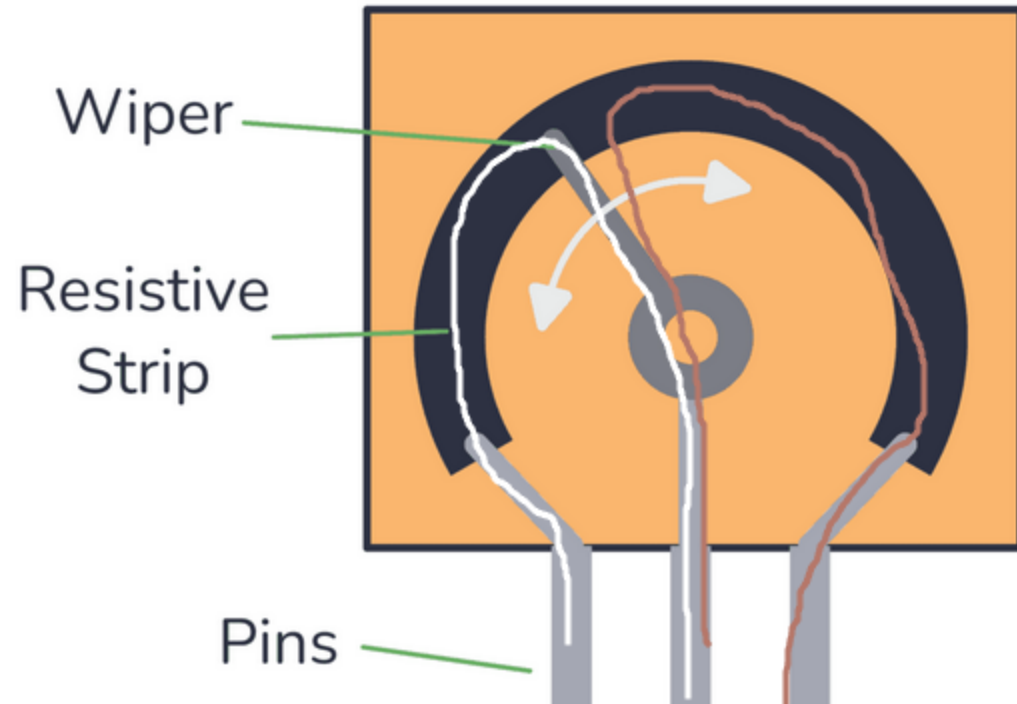
<https://wokwi.com/projects/376191892088461313>



- Was there increment in LED Glow?

Potentiometer : Variabe Resitor

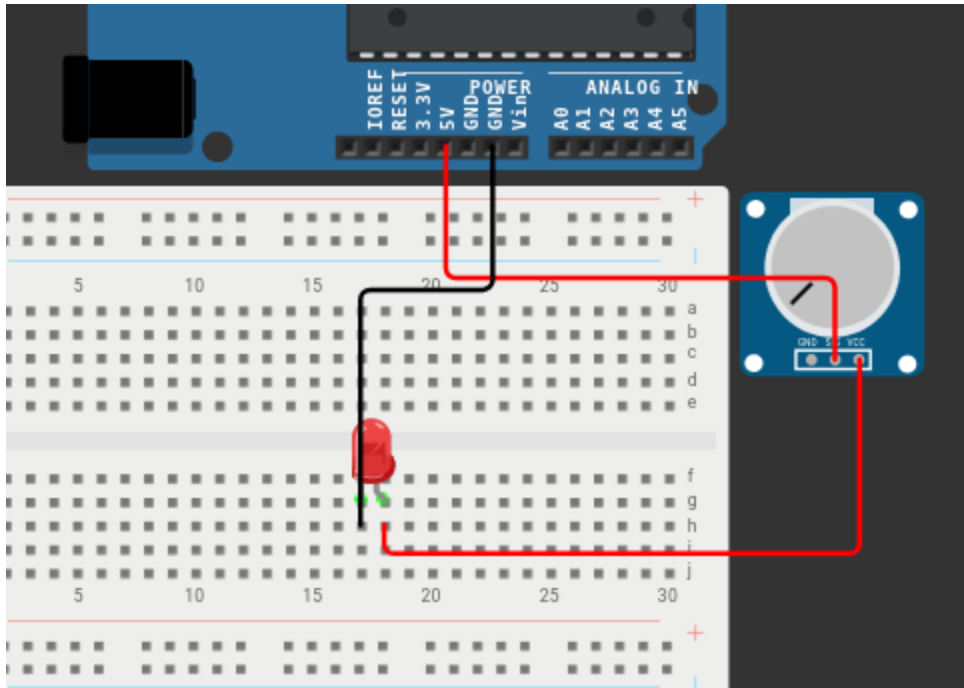
- $R = \rho \frac{L}{A}$



Build a circuit

- Potentiometer

<https://wokwi.com/projects/376209822976702465>



Installing Arduino IDE

- For Linux : <https://docs.arduino.cc/software/ide-v1/tutorials/Linux>
- For Windows : <https://docs.arduino.cc/software/ide-v1/tutorials/Windows>
- For Mac : <https://www.arduino.cc/en/Guide/macOS>

Overview of writing code on Arduino IDE

- Write code in Arduino IDE
- Connect Arduino to PC using USB cable
- Select the Port
- Compile the Program
- click button on IDE to load the program
- Done

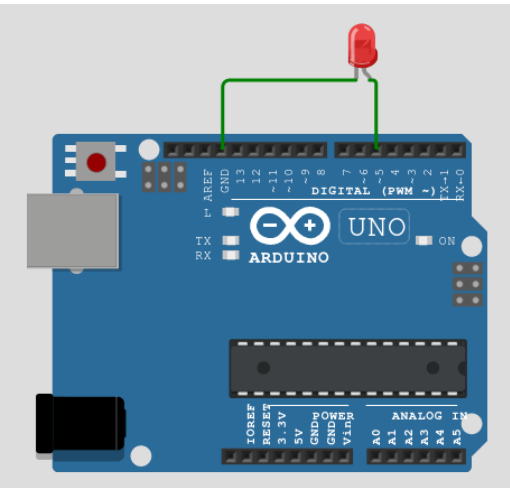
Write Some Code on Arduino IDE

General Structure

```
void setup() {  
    // setup code that runs once  
}  
  
void loop() {  
    // main code that runs repeatedly  
}
```


Example Code

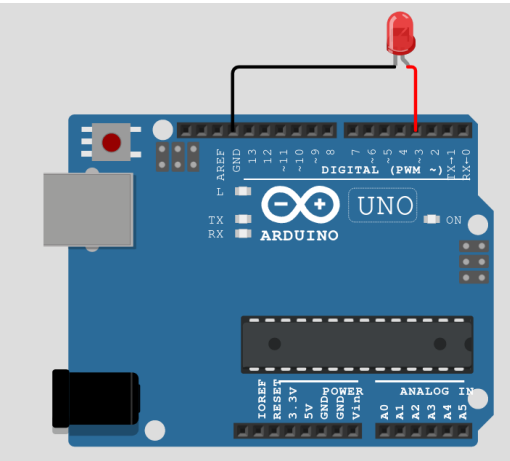
Link to Simulator: <https://wokwi.com/projects/338430633664578130>



```
void setup() {  
    pinMode(5,OUTPUT); //set pin no 5 on OUTPUT mode  
}  
  
void loop() {  
    digitalWrite(5, HIGH); // send 5V  
    delay(1000); // wait for 1 sec  
    digitalWrite(5,LOW); // send 0 volts  
    delay(1000); // wait for 1 sec  
}
```

Example Code B

Link to Simulator: <https://wokwi.com/projects/338430886227739218>



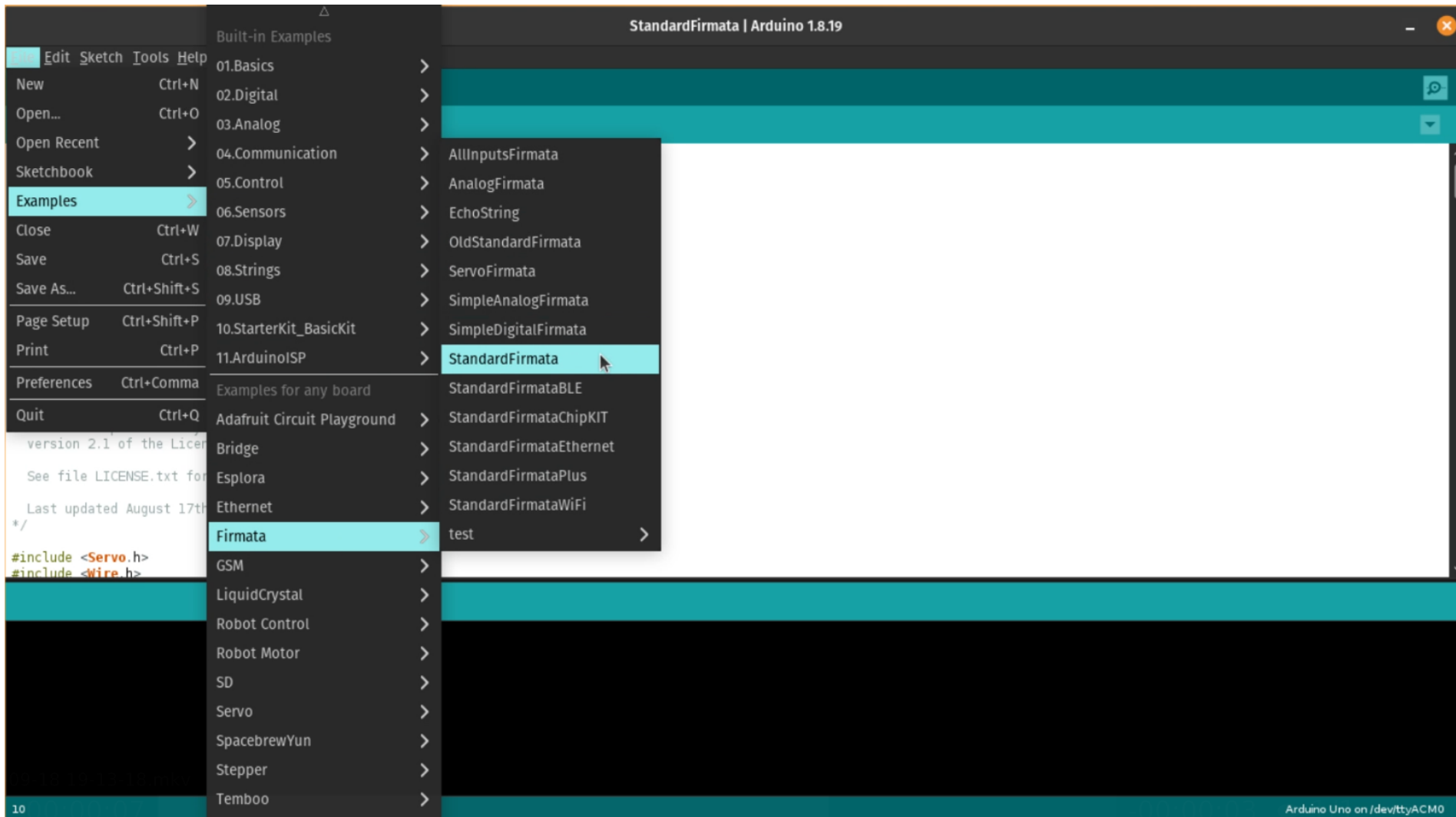
```
void setup() {  
  pinMode(3, OUTPUT); // connected to PWM output  
}  
  
void loop() {  
  analogWrite(3,255); // to show how bright LED is at first  
  delay(1000); // later fades away  
  
  for(int i=0;i<255;i++){  
    analogWrite(3,i); // analogWrite(pin, value) ; where value can range from 0 to 255.  
    delay(100); // 0.1 sec delay  
  }  
}
```

Firmata

- Protocol for communicating with microcontrollers from software on a computer.
- All you need is to implement the protocol then any microcontroller can be supported.

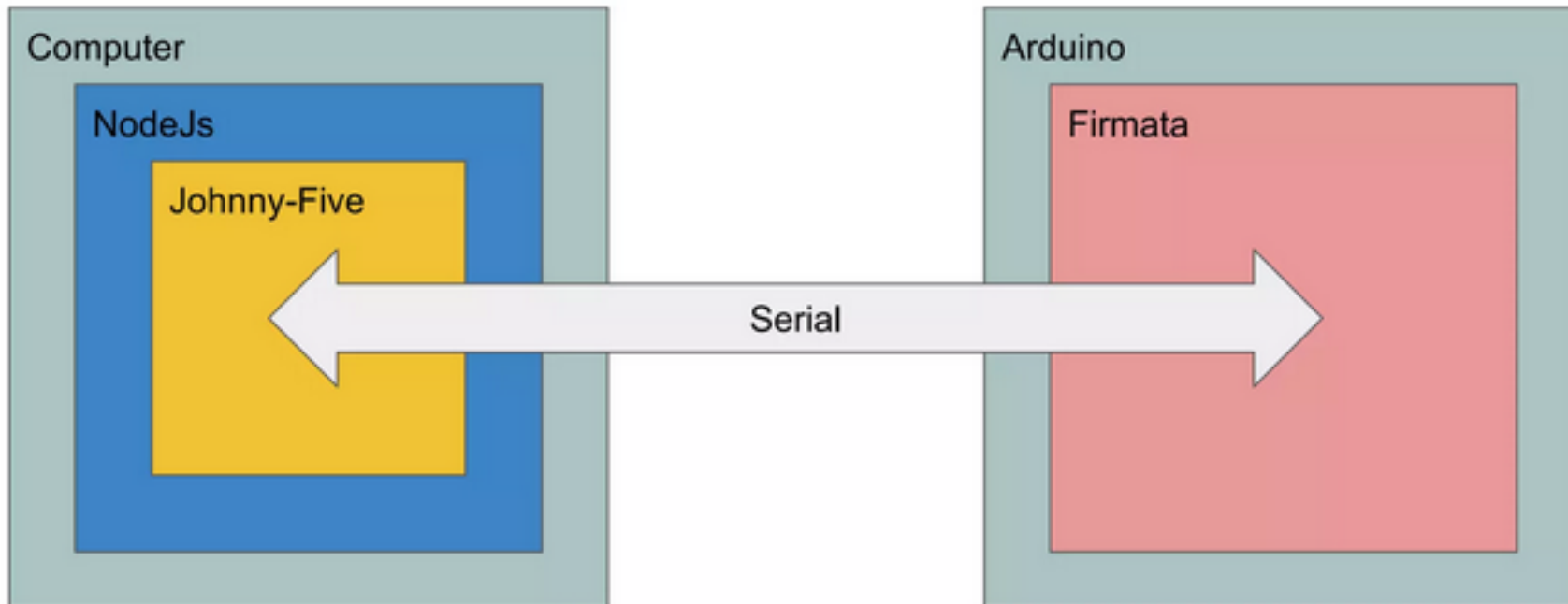
To Upload Firmata

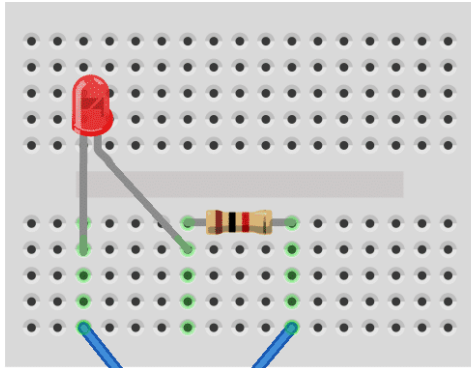
- connect the arduino to pc using USB cable
- open the arduino ide, select file>examples>firmata>StandardFirmata
- select port
- click upload



Firmata

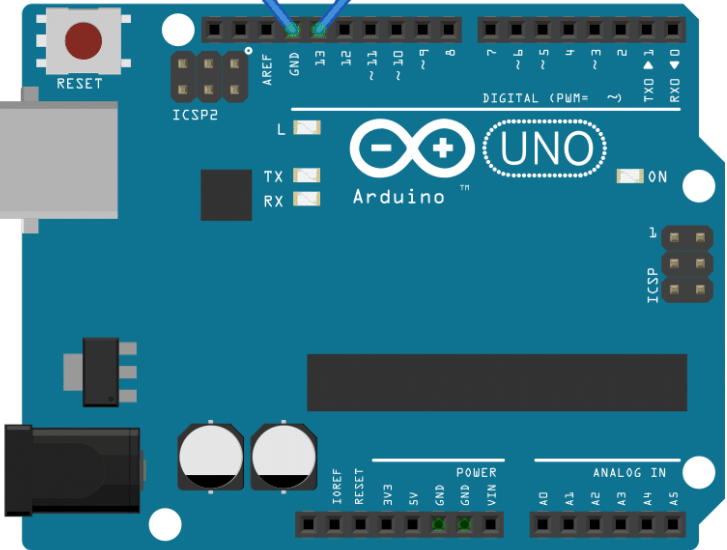
Basic Usage



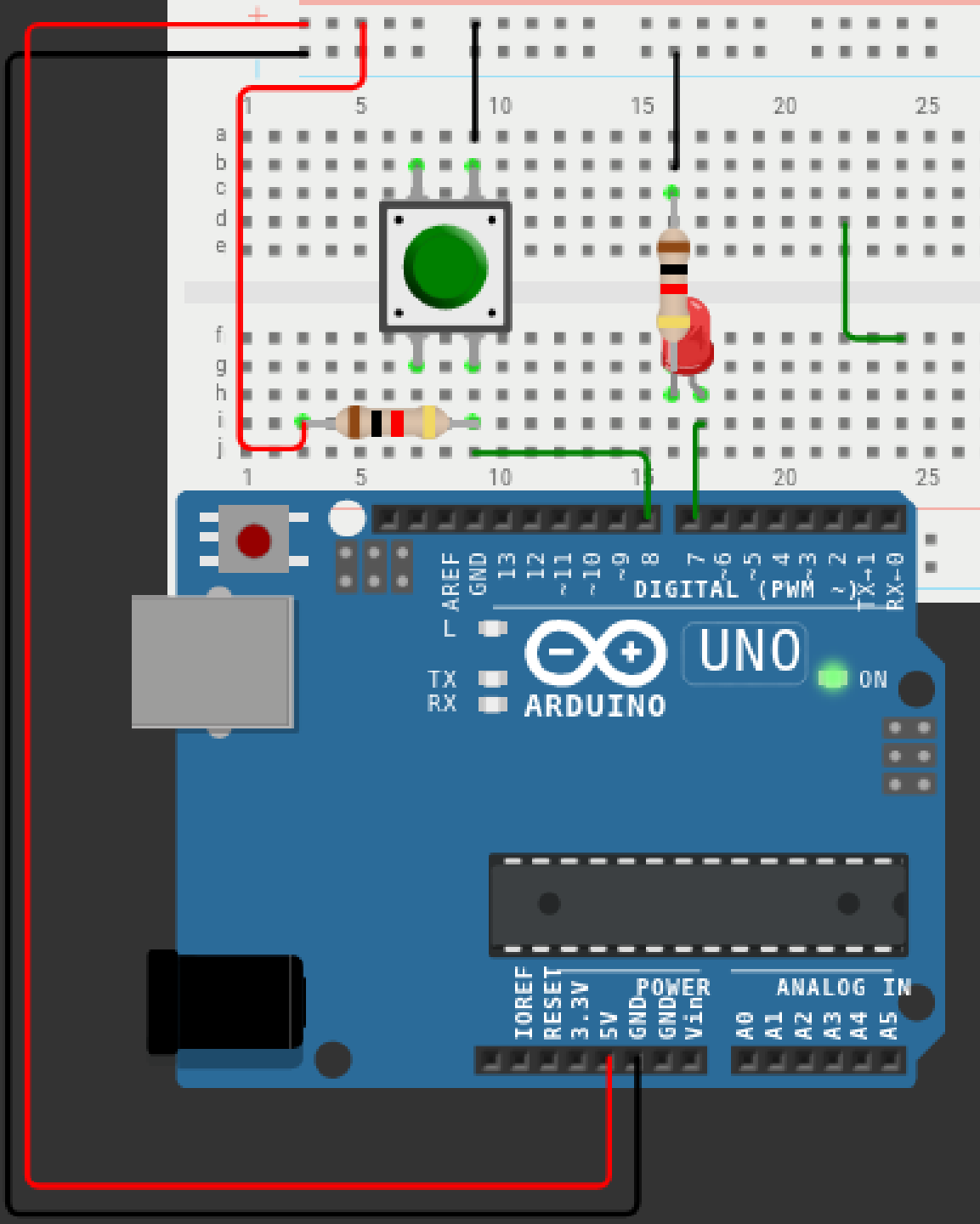


Led Blink

```
var five = require("johnny-five");  
var board = new five.Board();  
  
board.on("ready", function() {  
  var led = new five.Led(13);  
  led.blink(500);  
});
```



fritzing



PushButton

```
var five = require("johnny-five");
var board = new five.Board();

board.on("ready", function() {
  const led = new five.Led(7);
  const pushButton = new five.Button(
    {
      pin: 8,
      isPullup : true
    }
  );
  pushButton.on('down', () => {
    if (!led.isOn) {led.on();} else {led.off()}
  });
});
```

Analog Sensors

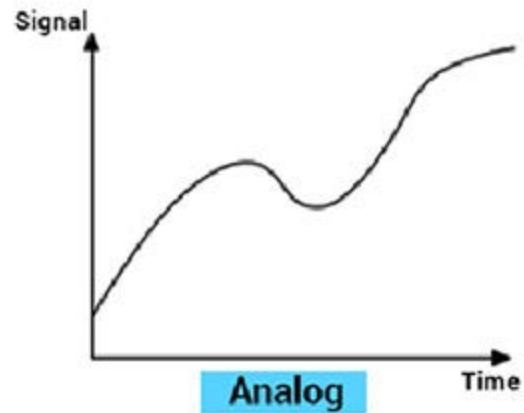
Analog Vs Digital



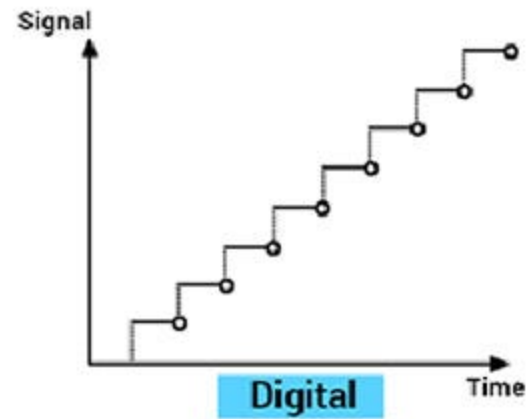
Analog signal



Digital signal



Analog

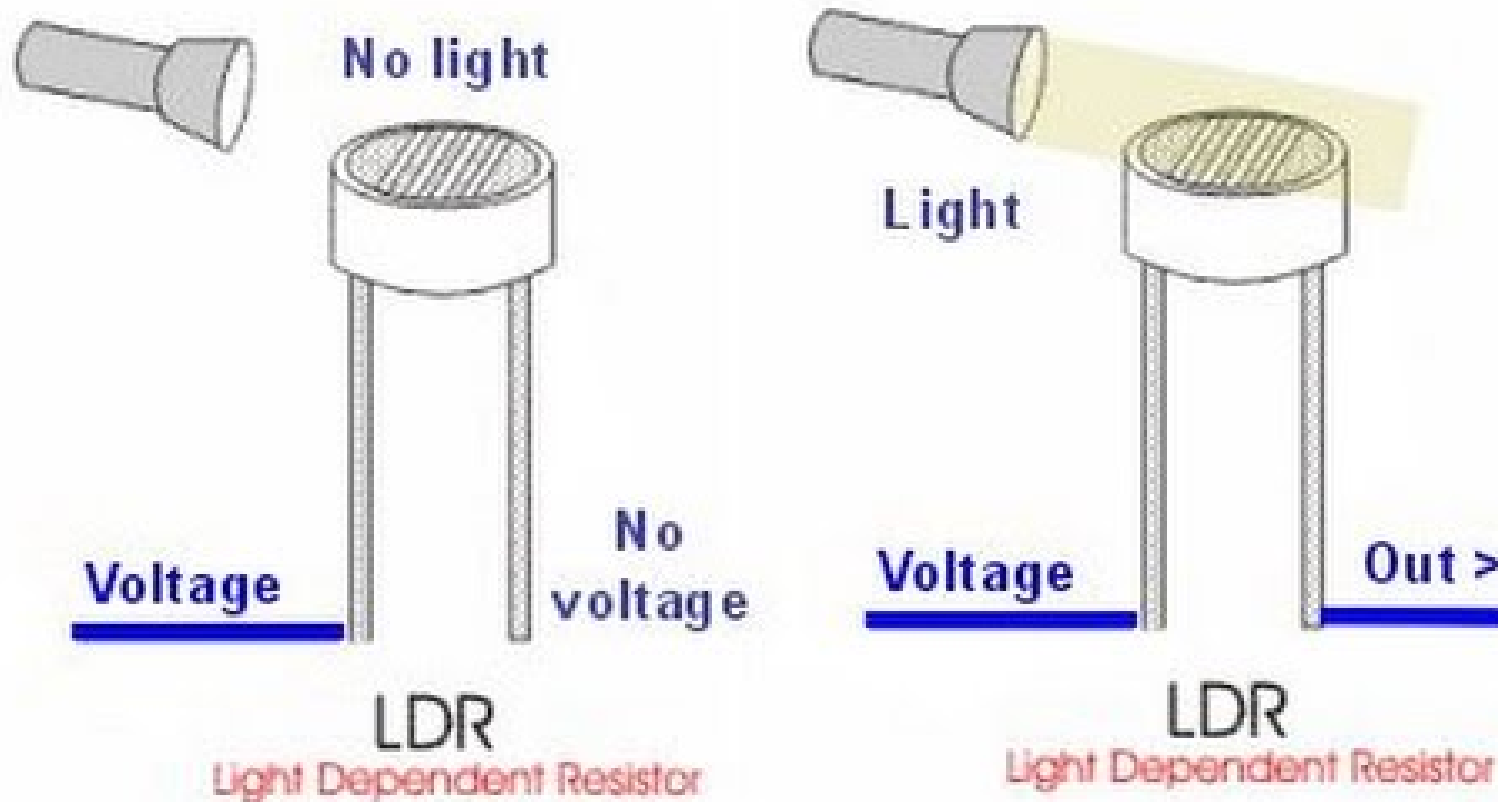


Digital

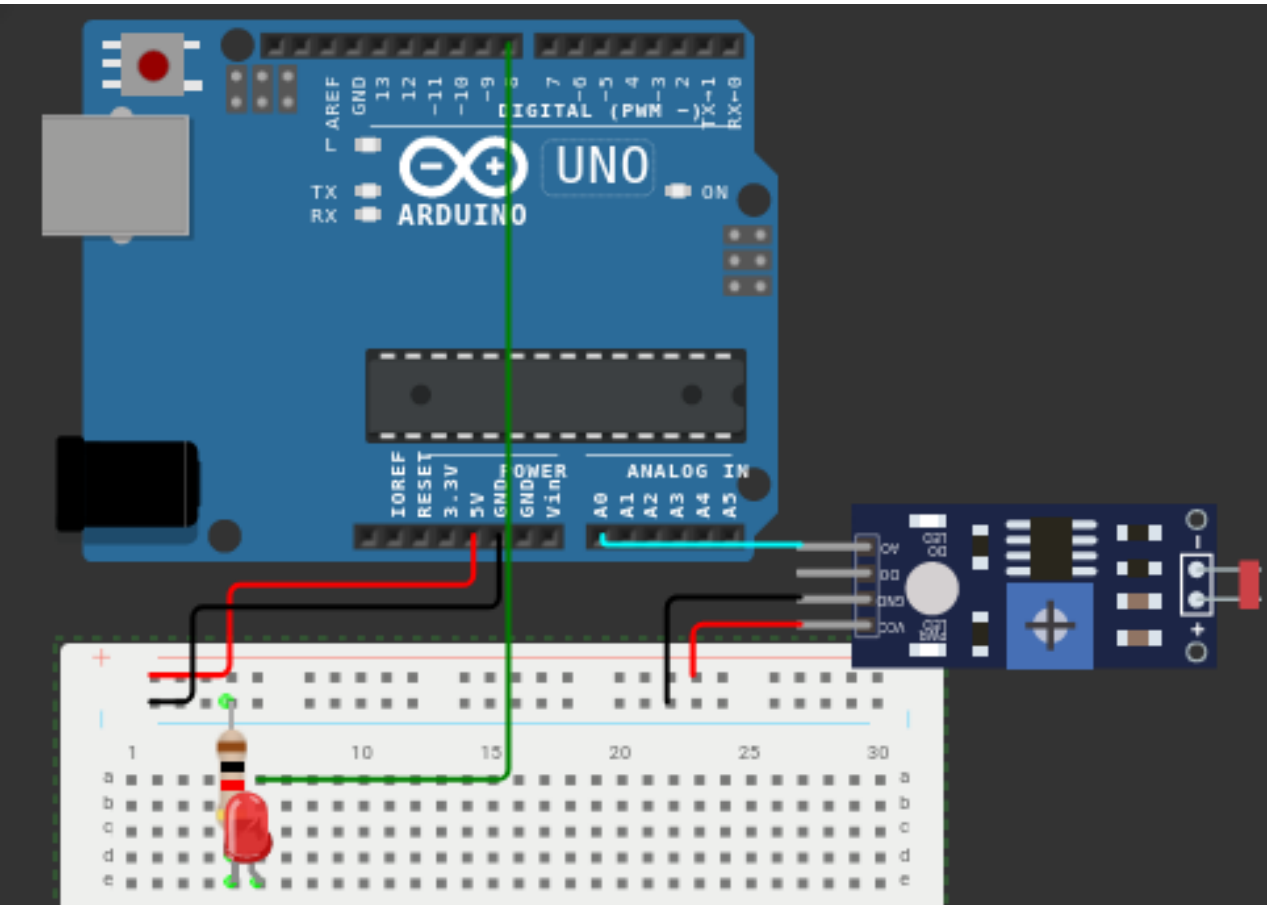
Analog Vs Digital

Aspect	Analog	Digital
Data Representation	Continuous waveform	Discrete binary values (0s and 1s)
Accuracy	High accuracy and precision	Excellent accuracy within bit resolution
Noise Resistance	Susceptible to noise and interference	More robust against noise
Signal Processing	Suitable for amplification, filtering, etc.	Versatile for complex algorithms
Cost and Complexity	Often simpler and potentially cheaper	Can be more complex and costly

LDR (Light Dependent Resistor)



<https://johnny-five.io/api/sensor/>



```
var five = require("johnny-five");

board = new five.Board();

board.on("ready", function() {

  let photoresistor = new five.Sensor({
    pin: "A0",
    freq: 500
  });

  let led = new five.Led(8);

  // board.repl.inject({
  //   pot: photoresistor
  // });

  photoresistor.on("data", function() {
    console.log(this.value);
    if (this.value >= 1000) {
      led.on();
    } else {
      led.off();
    }
  });

});
```

Equivalent C code

<https://wokwi.com/projects/376361780745305089>

```
#define LDR A0
#define LED 8

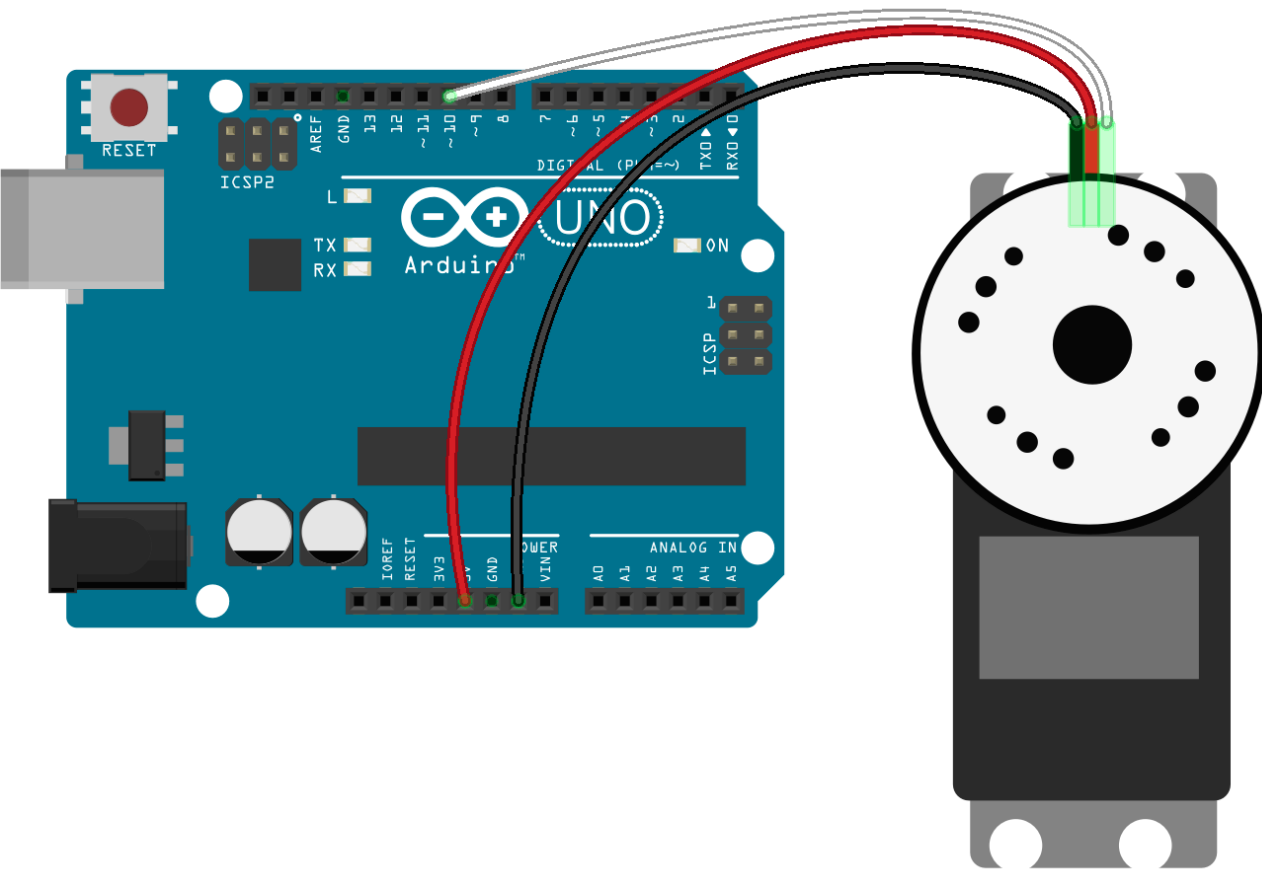
void setup() {
  pinMode(8, OUTPUT);
  pinMode(A0, INPUT);
}

void loop() {
  if (analogRead(A0) >= 1000) {
    digitalWrite(LED, HIGH);
  } else {
    digitalWrite(LED, LOW);
  }
}
```

Servo Control

Motors

Motor Type	Description	Image
Stepper	Converts electrical pulses into precise mechanical steps. Used in 3D printers and CNC machines.	
Servo	Provides accurate angular control, common in robotics.	
DC	Operates on direct current, simple, and used in various applications.	



Servo Motor

```
const five = require("johnny-five");
const board = new five.Board();

board.on("ready", () => {
  const servo = new five.Servo(10);

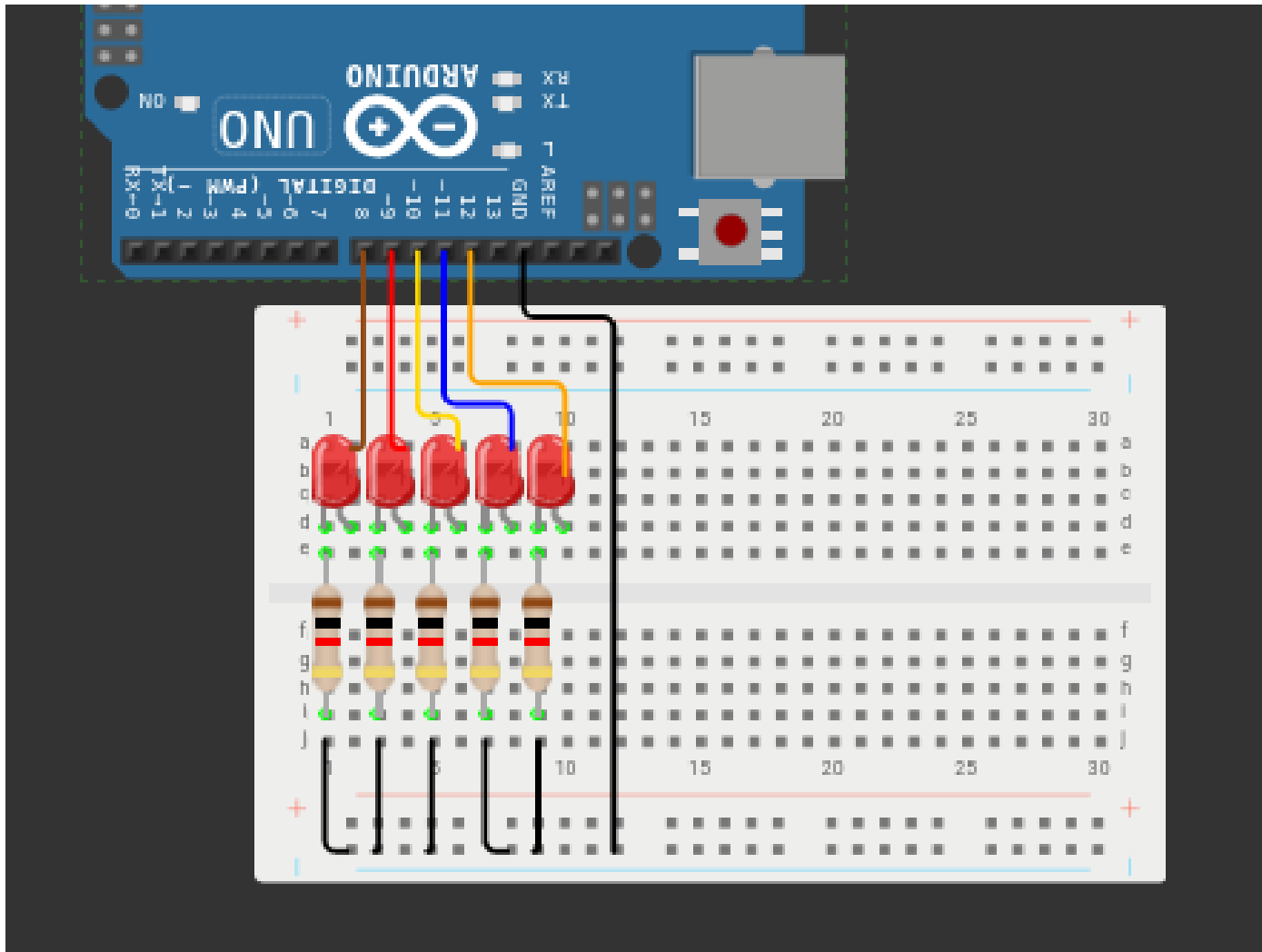
  board.repl.inject({
    servo
  });
  servo.sweep();
})
```


Build a project

LED Binary Counter

Circuit

<https://wokwi.com/projects/376380254565660673>



```

const five = require("johnny-five");
const board = new five.Board();

board.on("ready", () => {
  let leds = [8,9,10,11,12].map((pinNo) => {
    return new five.Led(pinNo);
  });

  for (let i = 0 ; i <= 31; i++) {
    setTimeout(() => {ledState(i.toString(2).padStart(5, '0') , leds)}, i * 1000);
  }

});

function ledState(ledString, leds) {
  Array.from("00000").map((val, index) => setLedState(leds[index], parseInt(val)));
  Array.from(ledString).map((val, index) => setLedState(leds[index], parseInt(val)));
}

function setLedState(led, state) {
  state == 1 ? led.on() : led.off();
}

```

Thankyou

